

An EM-based Detection Approach for Multiple Targets with Unknown Signal Signatures

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Abstract—In this paper, we address the problem of robust detection for multiple point-like targets in a background of Gaussian disturbance with unknown target signatures, including their range positions, angles of arrival, and number. Specifically, a joint detection and classification scheme is devised resorting to the likelihood ratio test in conjunction with the expectation-maximization algorithm within a hierarchical latent variable model. During the design phase, a batch of range bins are processed through a cyclic estimation procedure to integrate all the available backscattered energy from targets. Thereby, precise multiple targets parameter estimates are obtained via the maximum *a posteriori* rule. Finally, the superiority of the proposed architecture in terms of detection and classification performance is assessed through synthetic data experiments in comparison with conventional detectors.

Index Terms—Expectation-Maximization algorithm, hierarchical latent variable model, multiple targets detection, radar system, unknown target signature.

I. INTRODUCTION

In radar systems, adaptive target detection without any *a priori* information has consistently been a crucial task and drawn significant attention due to its importance [1]–[3]. In particular, this uncertainty spans several key characteristics of the targets, such as their complex amplitudes, their numbers, or their Angles of Arrival (AoAs), leading to profound implications for the detection algorithms. For example, most of the traditional space-time adaptive detection approaches, including the Generalized Likelihood Ratio Test (GLRT) [4], the Rao test [5], [6], and the two-step GLRT test [7], are derived for a single point-like target under the assumption of perfect alignment between the presumed and actual signal steering vectors. However, these tests might suffer significant performance degradation under conditions of AoA mismatch and/or in multiple targets scenarios. To address this issue, various robust detectors have been proposed to enhance their performance in the context of signal uncertainty.

In the open literatures, the subspace-based model is a conventional strategy for addressing angle uncertainty. This model is referred to the principle of invariance [8]–[11], and improves target discrimination by forming a signal subspace spanned by multiple steering vectors across the angular region of interest. For example, in [12], a GLRT-based detector is

derived in the homogeneous clutter, with the unknown steering vector residing within a known linear subspace. In [13], the authors extend this framework to partially homogeneous environment. To mitigate the target energy loss scattered across contiguous range bins resulting from AoA mismatch, several subspace-based detectors are proposed for range-distributed targets in [1], [14], [15]. Furthermore, the second order cone programming is an alternative approach to enhance detection robustness under uncertain AoAs [16]–[18]. However, these detectors do not return any estimates of the actual AoA and/or targets number. To address the problem in terms of AoA estimation, numerous sparse-decomposition-based methods have been developed [19]–[21]. Nevertheless, most of them are designed for either a point-like target or a range-spread target, and also depend on a sparsity-promoting prior knowledge [22].

In [23], GLRT-based detection schemes are introduced for extended and multiple point-like targets without assignment of secondary data. However, this method provides neither target enumeration nor AoA estimation. The work in [24] makes a further move on robust detection for multiple targets. Specifically, a novel LRT-based architecture is proposed assisted by the Expectation-Maximization (EM)-based estimation procedure. Again, this method assumes that multiple targets echoes originate from the same known direction, although enables estimation for targets number and range position.

In this paper, we take a further step toward the problem of multiple point-like targets detection with unknown signal signatures. To address this issue, we exploit the EM based estimation procedure combined with the hierarchical Latent Variable Model (LVM). Unlike the EM-based detector in [24], this design exploits a modified distribution that accounts for the two sets of hidden random variables at two different hierarchical levels in range-angle domain, thereby yielding new estimation updates and optimization procedures. Finally, a novel LRT-based detector is obtained by using the closed-form updates from the maximum *a posteriori* rule. The performance analysis indicates that the newly proposed detector demonstrates superior capability in signature estimation and multiple targets detection with respect to the considered competitors.

II. PROBLEM FORMULATION

Consider a scenario where multiple point-like targets are present and a radar system equipped with an array of N spatial channels senses the region of interest. Following the preliminary processing of the received echoes, K Independent and Identically Distributed (IID) vectors are collected across the range cells. Denote by $\mathbf{r}_k, k \in \Omega = \{1, \dots, K\}$, the N -dimensional random vector corresponding to the k th cell. The detection problem is to distinguish between the signal-plus-noise hypothesis (H_1) and noise-only hypothesis (H_0).

- Under H_0 , $\mathbf{r}_k$ are characterized as Circularly Symmetric Complex Gaussian (CSCG) random vectors with zero-mean and unknown inference covariance matrix $\mathbf{M}$, i.e., $\mathbf{r}_k \sim \mathcal{CN}(\mathbf{0}, \mathbf{M}), k \in \Omega$.
- Under H_1 , assume that $\Omega_T = \{k_1, \dots, k_T\}$ denotes the index subset of the range cells that contain target components, while $\Omega_C = \Omega \setminus \Omega_T$ indexes the range cells free of target. Notice that the main task is the preliminary detection of multiple targets without any *a priori* knowledge. Thus, $\mathbf{r}_k, k \in \Omega_T$ can be modeled as CSCG random vectors with unknown targets components and interference covariance matrix, while $\mathbf{r}_k, k \in \Omega_C$ are IID with the samples under H_0 . We have $\mathbf{r}_k \sim \mathcal{CN}(\alpha_k \mathbf{s}(\theta_k), \mathbf{M})$ for $k \in \Omega_T$, and $\mathbf{r}_k \sim \mathcal{CN}(\mathbf{0}, \mathbf{M})$ for $k \in \Omega_C$, with α_k and θ_k the unknown complex amplitudes and AoAs of targets, $\mathbf{s}(\theta_k)$ the N -dimensional nominal steering vector.

In the following, we address this problem through a hierarchical LVM that extends the standard LVM to a nested structure in range-angle bin. More precisely, let us first introduce hidden random variables a_k for each range cell to describe the presence of a possible target with unknown Probability Mass Function (PMF) $P(a_k = s) = \pi_s$ following the constraint $\sum_{s=0}^1 \pi_s = 1$. Then, we formulate a hierarchical targets-present case (namely $s = 1$) with an inner LVM. Before this, denote by $\Theta = \{\theta_1, \dots, \theta_L\}$ the discrete and finite set of angles within surveillance. To describe the unknown AoA at the k th range, we assume IID random variables $b_k, k = 1, \dots, K$ satisfy the unknown PMF $P(b_k = \theta_l) = p_l, l = 1, \dots, L$ and $\sum_{l=1}^L p_l = 1$. Summarizing, the detection problem of interest can be formulated as the following test

$$\begin{cases} H_0 : \mathbf{r}_k \sim \mathcal{CN}(\mathbf{0}, \mathbf{M}), k \in \Omega, \\ H_1 : \begin{cases} \mathbf{r}_k | a_k = 0 \sim \mathcal{CN}(\mathbf{0}, \mathbf{M}), k \in \Omega_C, \\ \mathbf{r}_k | b_k = \theta_l, a_k = 1 \sim \mathcal{CN}(\alpha_{k,l} \mathbf{s}(\theta_l), \mathbf{M}), k \in \Omega_T. \end{cases} \end{cases} \quad (1)$$

III. ARCHITECTURE DESIGNS

To deal with (1), we devise a novel multi-target detection approach resorting to the LRT criterion. Let $\mathbf{R} = [\mathbf{r}_1, \dots, \mathbf{r}_K]$ denote the overall data matrix, with logarithm of the joint Probability Density Function (PDF) given by $\mathcal{L}(\mathbf{R}; \mathcal{P}) = \sum_{k=1}^K \log f(\mathbf{r}_k; \mathcal{P}_s)$, where $\mathcal{P}_s = [s\alpha_{k,l}, s\theta_l, sT, s\Omega_T, \mathbf{M}]$ presents the set of the unknown parameters under $H_s, s = 0, 1$

hypothesis, $\mathcal{P} = \{\cup_{k=1}^K \mathcal{P}_s\} \cup \{\pi_s, p_l\}$, $f(\mathbf{r}_k; \mathcal{P}_s)$ is the PDF of $\mathbf{r}_k$ and given by

$$f(\mathbf{r}_k; \mathcal{P}_s) = \pi_0 f(\mathbf{r}_k | a_k = 0; \mathcal{P}_0) + \pi_1 \sum_{l=1}^L f(\mathbf{r}_k | b_k = l, a_k = 1; \mathcal{P}_1) p_l, \quad (2)$$

with

$$f(\mathbf{r}_k | sb_k = l, a_k = s; \mathcal{P}_s) = e^{-\text{Tr}[\mathbf{M}^{-1}(\mathbf{r}_k - s\alpha_{k,l} \mathbf{s}(\theta_l))(\mathbf{r}_k - s\alpha_{k,l} \mathbf{s}(\theta_l))^\dagger]} / \pi^N |\mathbf{M}| \quad (3)$$

the conditional PDFs given $a_k = 0$ (for $s = 0$) and $b_k = l, a_k = 1$ (for $s = 1$), respectively. For future reference, symbols $|\cdot|$, $\text{Tr}(\cdot)$, $(\cdot)^\dagger$ and $(\cdot)^{-1}$ denote the determinant, trace, conjugate transpose, and inverse. Within the framework of LRT, the adaptive detector can be written as

$$\prod_{k=1}^K f(\mathbf{r}_k; \hat{\mathcal{P}}_1) / f(\mathbf{r}_k; \hat{\mathcal{P}}_0) \stackrel{H_1}{\underset{H_0}{\gtrless}} \eta, \quad (4)$$

where η is the detection threshold set according to the required Probability of False Alarm (P_{fa}), $\hat{\mathcal{P}}_1$ is the final estimates provided by the following EM-based optimization procedure and $\hat{\mathcal{P}}_0 = \{\hat{\mathbf{M}} = \mathbf{R}\mathbf{R}^\dagger / K\}$ is the maximum likelihood estimate of $\mathbf{M}$ under H_0 [25]–[27]. Now, we proceed with the E-step and the M-step under H_1 .

A. E-step

Exploiting the Jensen's Inequality [28], [29], the E-step is tantamount to implementing the update rule

$$q_k^{(m-1)}(s) = \frac{F_s^{(m-1)} \hat{\pi}_s^{(m-1)} e^{-u(s, \rho)}}{F_0^{(m-1)} \hat{\pi}_0^{(m-1)} e^{-u(0, \rho)} + F_1^{(m-1)} \hat{\pi}_1^{(m-1)} e^{-u(1, \rho)}}, s = 0, 1, \\ r_k^{(m-1)}(l) = \frac{f(\mathbf{r}_k | b_k = l, a_k = 1; \hat{\mathcal{P}}_1^{(m-1)}) \hat{p}_l^{(m-1)}}{F_1^{(m-1)}}. \quad (5)$$

In (5), $F_0^{(m-1)} = f(\mathbf{r}_k | a_k = 0; \hat{\mathcal{P}}_0^{(m-1)})$, $F_1^{(m-1)} = \sum_{l=1}^L f(\mathbf{r}_k | b_k = l, a_k = 1; \hat{\mathcal{P}}_1^{(m-1)}) \hat{p}_l^{(m-1)}$, $e^{-u(s, \rho)} = e^{-(N^2 + 3s)(1 + \rho)/2}$, $s = 0, 1, \rho \geq 1$, is the penalty term inherited from the Generalized Information Criterion (GIC) [24], [30] to mitigate the overestimation of target components, $\hat{\mathcal{P}}_s^{(m-1)}$, $\hat{\pi}_s^{(m-1)}$, and $\hat{p}_l^{(m-1)}$ represent the estimates of the corresponding parameters at the $(m-1)$ th iteration.

B. M-step

In the M-step, we obtain the estimates of $\mathcal{P}_1$ by maximizing the following problem

$$\max_{\mathcal{P}_1} \sum_{k=1}^K \left\{ q_k^{(m-1)}(0) \log f(\mathbf{r}_k | a_k = 0; \mathcal{P}_0) \pi_0 + q_k^{(m-1)}(1) \log \pi_1 + q_k^{(m-1)}(1) \sum_{l=1}^L r_k^{(m-1)}(l) \log f(\mathbf{r}_k | b_k = l, a_k = 1; \mathcal{P}_1) p_l \right\}. \quad (6)$$

Notice that the PMF values are independent of the rest of parameters in $\mathcal{P}_1$. Therefore, we first conduct the maximization with respect to π_s and p_l by using the Lagrange multipliers' method and have [24], [31]

$$\begin{aligned}\hat{\pi}_s^{(m)} &= \sum_{k=1}^K q_k^{(m-1)}(s)/K, \quad s = 0, 1, \\ \hat{p}_l^{(m)} &= \frac{\sum_{k=1}^K q_k^{(m-1)}(1)r_k^{(m-1)}(l)}{\sum_{k=1}^K q_k^{(m-1)}(1)}, \quad l = 1, \dots, L.\end{aligned}\quad (7)$$

Then, plugging (3) into (6) and ignoring the irrelevant terms, the maximization problem with respect to $\alpha_{k,l}$ and $\mathbf{M}$ can be recast as

$$\begin{aligned}\max_{\substack{\mathbf{M}, \alpha_{k,l} \\ k \in \Omega, l=1, \dots, L}} & -K \left\{ \log |\mathbf{M}| + \text{Tr} \left[\mathbf{M}^{-1} \sum_{k=1}^K \left(q_k^{(m-1)}(0) \mathbf{S}_k \right. \right. \right. \\ & \left. \left. \left. + \sum_{l=1}^L q_k^{(m-1)}(1) r_k^{(m-1)}(l) \mathbf{S}_{l,k} \right) / K \right] \right\},\end{aligned}\quad (8)$$

with $\mathbf{S}_k = \mathbf{r}_k \mathbf{r}_k^\dagger$, $\mathbf{S}_{l,k} = (\mathbf{r}_k - \alpha_{k,l} \mathbf{s}(\theta_l))(\mathbf{r}_k - \alpha_{k,l} \mathbf{s}(\theta_l))^\dagger$. Let us exploit the inequality $\log |\mathbf{A}| - \text{Tr} \mathbf{A} \leq -N$ for $\mathbf{A}$ any N -dimensional matrix with nonnegative eigenvalues [32], [33]. Hence, the estimate update for $\mathbf{M}$ is

$$\begin{aligned}\widehat{\mathbf{M}}^{(m)} &= \\ \frac{1}{K} \sum_{k=1}^K & \left(q_k^{(m-1)}(0) \mathbf{S}_k + q_k^{(m-1)}(1) \sum_{l=1}^L r_k^{(m-1)}(l) \mathbf{S}_{l,k} \right).\end{aligned}\quad (9)$$

Replacing $\mathbf{M}$ with $\widehat{\mathbf{M}}^{(m)}$ and ignoring the terms independent of $\alpha_{k,l}$, (8) is tantamount to

$$\min_{\substack{\alpha_{k,l} \\ k \in \Omega, l=1, \dots, L}} K \log \left| \frac{\mathbf{E}_l^{(m-1)} + q_k^{(m-1)}(1) \sum_{l=1}^L r_k^{(m-1)}(l) \mathbf{S}_{l,k}}{K} \right|,\quad (10)$$

with $\mathbf{E}_l^{(m-1)} = \sum_{k=1}^K \left\{ q_k^{(m-1)}(1) \sum_{l'=1, l' \neq l}^L r_k^{(m-1)}(l') \mathbf{S}_{l',k} + q_k^{(m-1)}(0) \mathbf{S}_k \right\}$. In (10), we introduce an iterative maximization procedure with respect to $\bar{\alpha}_l = [\alpha_{1,l}, \dots, \alpha_{K,l}]^T$. Specifically, we assume that $\bar{\alpha}_i, i \neq l$, are known and maximize the objective function respect to $\bar{\alpha}_l$. The final update can be written as

$$\begin{aligned}(\hat{\bar{\alpha}}_l^T)^{(m)} &= \left[\frac{\mathbf{s}(\theta_l)^\dagger (\mathbf{E}_l^{(m-1)})^{-1} \mathbf{X}_l}{\mathbf{s}(\theta_l)^\dagger (\mathbf{E}_l^{(m-1)})^{-1} \mathbf{s}(\theta_l)} \right] \\ &\times \text{diag} \left\{ (Q_{1,l}^{(m-1)})^{-1/2}, \dots, (Q_{K,l}^{(m-1)})^{-1/2} \right\},\end{aligned}\quad (11)$$

where $Q_{1,l}^{(m-1)} = q_k^{(m-1)}(1) r_k^{(m-1)}(l) > 0$, and $\mathbf{X}_l = \left[\sqrt{Q_{1,l}^{(m-1)}} \mathbf{r}_1, \dots, \sqrt{Q_{K,l}^{(m-1)}} \mathbf{r}_K \right]$ with $k \in \Omega, l = 1, \dots, L$. The estimation procedure is summarized in Algorithm 1.

Algorithm 1: Iterative Estimation Procedure.

for m **do**

E-step: update rules $q_k^{(m-1)}(s)$ and $r_k^{(m-1)}(l)$;

M-step:

- update for $\hat{\pi}_s^{(m)}$ and $\hat{p}_l^{(m)}$;
- estimate the $(\hat{\bar{\alpha}}_l^T)^{(m)}$ by an inner iterative procedure;
- estimate $\widehat{\mathbf{M}}^{(m)}$ with known $\hat{\bar{\alpha}}_{k,l}^{(m)}$.

end

C. Convergence of the EM-based Procedure

The EM-based procedure continues until the following stopping criterion is not satisfied,

$$\Delta \mathcal{G}(m) = \frac{|\mathcal{L}(\mathbf{R}; \widehat{\mathcal{P}}^{(m)}) - \mathcal{L}(\mathbf{R}; \widehat{\mathcal{P}}^{(m-1)})|}{|\mathcal{L}(\mathbf{R}; \widehat{\mathcal{P}}^{(m)})|} < \delta, \quad (12)$$

or $m = m_{max}$, where $\delta > 0$ and m_{max} is the maximum number of EM iterations, $|\cdot|$ denotes the modulus of a real number. Note that $\widehat{\Omega}_T = \{k \in \Omega : \max_{s=0,1} \hat{q}_k(s) = 1\}$ represents an estimate of the set of range bins containing target components, targets number equals the cardinality of set $\widehat{\Omega}_T$. Moreover, targets AoA can be indexed by exploiting the rule $\hat{l}_k = \max_{l=1, \dots, L} \hat{r}_k(l)$ with $k \in \widehat{\Omega}_T$. $\hat{q}_k(s)$ and $\hat{r}_k(l)$ are the final estimates of $q_k(s)$ and $r_k(l)$ ¹.

IV. PERFORMANCE ASSESSMENT

In this section, we analyze the performance of the proposed scheme based on synthetic data. Specifically, we present numerical examples assuming $N = 8$, $K = 24$, $\Theta = [-20^\circ : 2^\circ : 20^\circ]$ (namely $L = 21$), $\delta = 10^{-4}$, and $\rho = 3^2$. Three point-like targets appear in the 6th, 13th, and 16th range bin with the same power. The AoA of the targets are assumed to lie on the sampling grid and point at $\theta_1 = -16^\circ$, $\theta_2 = 4^\circ$ and $\theta_3 = 12^\circ$. The Signal-to-Interference-plus-Noise Ratio (SINR) of the i th target in the k th range bin is defined as $\text{SINR} = \|\alpha_{k,i}\|^2 \mathbf{s}(\theta_i)^\dagger \mathbf{R}^{-1} \mathbf{s}(\theta_i)$, with $\mathbf{s}(\theta_i) = [1, \dots, e^{-j\pi(N-1)\sin(\theta_i)}] / \sqrt{N}$ [35]. Finally, the EM algorithm is initialized with $\pi_s^{(0)} = 1/2$, $p_l^{(0)} = 1/L$, $\mathbf{M}^{(0)} = \mathbf{R} \mathbf{R}^\dagger / K$.

Firstly, we investigate the convergence property of the proposed EM-based procedure. In Fig. 1, the relative variation of $\mathcal{L}(\mathbf{R}; \mathcal{P})$ versus the SINR are plotted for various iteration numbers over 1000 Monte Carlo (MC) trials. The curves confirm that the relative variations in (12) are lower than 10^{-4} when $m \geq 4$. Furthermore, notice that the results in Fig.1 slightly decrease in the SINR range of 15 dB to 20 dB. This is due to the fact that the EM algorithm converges more rapidly as the target energy increases, resulting in lower variations

¹We assume only one target in each range bin, whose AoA is jointly estimated with the other parameters resorting to EM.

²Actually, The "optimal" value of ρ in the GIC rule depends on several aspects such as the performance metrics, the number of samples, and the data-generating mechanism itself [34, and references therein]. For this reason, this parameter is set by heuristically selecting the value that returns on average a reliable performance in all the considered parameter scenarios.

TABLE I
RMSES FOR THE AOA AND NUMBER OF TARGETS OVER 1000 TRIALS

SINR / dB	15	20	25	30
$\text{RMSE}_{\text{AoA}} / ^\circ$	5.7	0.8	0.3	0.1
RMSE_T	0.7	0.1	0.1	0.1

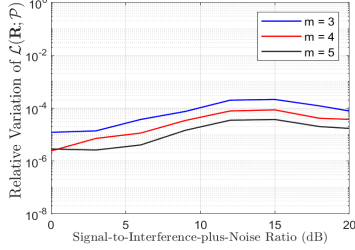


Fig. 1. Mean relative variations of $\mathcal{L}(\mathbf{R}; \mathcal{P})$ versus SINR for various m .

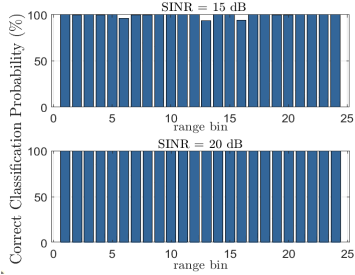


Fig. 2. CCP values (%) for target position resorting to 1000 MC.

for the same number of iterations. In Fig. 2, we analyze the Correct Classification Probability (CCP, %) of target positions resorting to 1000 MC trials. It can be observed that the proposed method achieves reliable classification performance with CCP values more than 90% for SINR = 15 dB and more than 99% for SINR = 20 dB. Fig. 3 shows the Probability of Classification (P_c , %) with respect to the AoAs of the three targets over 1000 MC. The results indicate that the percentage of selecting the true AoA values is greater than 70% when SINR = 15 dB and than 80% when SINR = 20 dB³. In Table I, we show the Root Mean Square Errors associated with the estimation of the AoAs (RMSE_{AoA}) and the number of targets (RMSE_T) over 1000 trials. It can be seen that the values of RMSE_{AoA} and RMSE_T are less than 1 when SINR $\geq$ 20 dB.

The last illustrative example provides an analysis of detection performance for the proposed method in comparison with the detector (4) in [23] and detectors (11), (16) in [24]. To ensure a fair comparison, the competitors in [24] consider only one homogeneous clutter region, and terminate when the relative variation of their objective function is less than 10^{-4} . Moreover, the nominal AoA used in the competitors is equal to

³It should be noted that detection is a necessary condition to trigger estimation. In Figs. 2-3, the SINR values are selected to guarantee a high probability of detection.

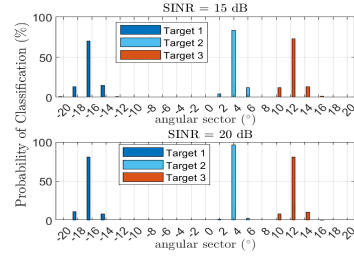


Fig. 3. P_c (%) for angular position resorting to 1000 MC.

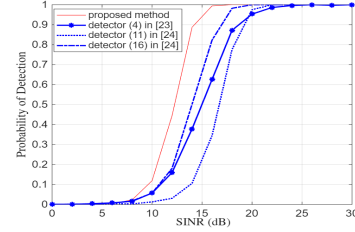


Fig. 4. P_d versus SINR for $P_{fa} = 10^{-4}$ over 1000 MC.

4° . In Fig. 4, the Probability of Detection (P_d) curves versus SINR are computed exploiting 1000 independent trials with the $P_{fa} = 10^{-4}$. Results show that the proposed detector experiences a superior detection performance compared to other competitors with a gain of no less than about 2.8 dB for $P_d = 0.9$. Notably, this improvement originates from the precise estimation of multiple targets parameters, whereas competing approaches suffer performance degradation as they are unable to fully capture the entire target energy owing to mismatches in AoA and/or range bins.

V. CONCLUSIONS

In this paper, a binary hypothesis test is first formulated by using a two-level hierarchical LVM that characterizes the probabilities of target presence and the associated pointing direction. Then, we apply the EM algorithm and devise an iterative estimation procedure to provide reliable estimates for the positions and the AoAs of the targets in the range-angle domain. Finally, a novel adaptive detector is proposed based on the LRT framework with the estimates obtained from the EM. Numerical examples conducted on synthetic data show that the EM-based approach exhibits a significant advantage with respect to the considered competitors. Possible extensions could concern the problem of robust detection of multiple point-like targets to heterogeneous scenarios or to non-Gaussian interference.

VI. ACKNOWLEDGMENT

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⁴Please note that the competitors assume that multiple point-like targets originate from one same known direction, which leads to inherent AoA mismatches and, inevitably, to a certain performance loss.

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